

Module 01: Systems — Digital and Physical Systems

Learning Objectives

1. Identify the **components** (hardware, software, network) of a digital system and map them to Active Inference concepts.
2. Compare **centralized** and **distributed** system architectures.
3. Explain how a computer's operating system manages resources by minimizing "surprise" (unexpected demand).

Introduction

Every piece of technology you use — your phone, a self-driving car, a social media platform — is a system. In Math Foundations, you learned to model systems with equations. Now we apply those ideas to the systems that define modern life: computers, networks, AI platforms, and the Internet itself.

Key Concepts

1. Anatomy of a Digital System

A smartphone is a layered system:

- **Hardware layer:** CPU, RAM, battery, sensors (camera, accelerometer, GPS)
- **Operating system layer:** Manages resources, schedules tasks, handles I/O
- **Application layer:** Apps that the user interacts with

Each layer has a **boundary** (like a Markov blanket). The OS mediates between hardware and apps — it "hides" the hardware complexity from the apps, just like a brain's Markov blanket hides the raw physics of the world.

2. Centralized vs. Distributed Systems

A **centralized** system (e.g., a single database server) has one decision-maker. A **distributed** system (e.g., the Internet, blockchain, a flock of birds) has many agents making local decisions.

Active Inference works for both: a single agent minimizes its own free energy, while a collective of agents can minimize *collective* free energy through communication and coordination.

3. Resource Prediction

An operating system predicts resource demand: "The user usually opens email at 8 AM, so pre-load it." When demand is unpredictable (surprise!), the system must adapt — allocating more RAM, throttling background processes, or requesting a restart. This is active inference applied to computer resource management.

Applications

- **Cloud Computing:** Amazon Web Services (AWS) uses predictive scaling — it *predicts* how many servers you will need tonight based on past traffic patterns and pre-allocates resources. Unexpected traffic spikes are prediction errors that trigger auto-scaling.
- **The Internet as a System:** The Internet has no central brain. Packets route themselves through millions of nodes using local rules. This is a massively distributed system performing something analogous to free energy minimization at a planetary scale.

Discussion Questions

1. Your phone battery dies unexpectedly at 3 PM instead of lasting until 6 PM. What "prediction error" did the phone's power management system make?
2. Is the Internet centralized or distributed? What are the advantages of each for resilience?

Summary

Digital systems have components, boundaries, and prediction mechanisms, just like biological systems. Operating systems and cloud platforms manage resources by predicting demand and correcting errors — the same predict-correct cycle that underlies Active Inference.

References

- Tanenbaum, A. S. (2016). *Modern Operating Systems*. Chapter 1.